

TASK 1

Project Definition, Clinical Tasks & Dataset Specification

Project: Badira: Profile-Guided Trustworthy Survival Intelligence for Women's Health

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1. What the Project Is — Master Definition

BADIRA is a self-adaptive framework for predicting the risk of pregnancy complications.

BADIRA first creates a standardized Woman Schema that organizes relevant information about each woman, such as age, nationality, pregnancy history, family history, and clinical measurements.

Based on the Woman Schema, BADIRA collects the required data and uses the Data Profiler to assess the quality and characteristics of the training data, including missing values, imbalance, distribution, and subgroup coverage.

BADIRA then uses this information to select the most suitable model and prediction settings. The model predicts the woman's risk over time and provides a confidence score and explanation.

The goal of BADIRA is to provide predictions that are accurate, trustworthy, understandable, and adaptable to different data conditions.

BADIRA will first be tested on preeclampsia and later extended to preterm birth and gestational diabetes (GDM).

2. BADIRA Architecture / Pipeline



3. The Three Clinical Tasks

BADIRA will be tested on **three pregnancy complications**. The same framework will be used for all three, while the data and prediction target change.

3.1 Task Comparison Table

	Preeclampsia	Preterm Birth	GDM
What we predict	Risk of developing preeclampsia	Risk of giving birth too early	Risk of developing gestational diabetes
Prediction time	11–14 weeks	Early pregnancy	Early pregnancy
Outcome	Preeclampsia during pregnancy	Birth before 37 weeks	GDM during pregnancy
Main data used	Blood pressure, BMI, age, parity, biomarkers	Previous preterm birth, cervical length, BMI, smoking	BMI, age, family history, previous GDM

4. Profiling Requirements

4.1 Data Profiler

Profiling Dimension	Description
Imbalance Signature	Checks how common the pregnancy complication is and whether the dataset has a strong imbalance between women with and without the complication.
Missingness Mechanism	Examines the pattern of missing data and whether missing information may affect the reliability of the model.
Shift Vulnerability	Checks whether the data distribution may be different across patient groups, locations, or time periods.

Temporal Structure	Look at how information is collected over time, including visit timing and changes in measurements.
Dataset Scale	Evaluates the number of women, number of complication cases, and amount of available data to determine whether the dataset is large enough for the model.
Threat / Perturbation Profile	Identifies possible weaknesses in the data and the types of changes or disturbances the model may need to handle.

4.2 Woman Profiler

Profiling Dimension	Description
Basic Information	Age, nationality
Pregnancy History	Number of previous pregnancies, previous births
Previous Complications	Previous preeclampsia, previous preterm birth, previous GDM
Health History	Chronic hypertension, diabetes, kidney disease, other relevant conditions
Current Pregnancy Information	BMI, blood pressure, pregnancy type, and other available information
Risk Background	Whether the woman has known factors that may increase her risk

5. Data Separation Rule

BADIRA will use the training data to build the profiles and make model-selection/configuration decisions. The test data will remain untouched until the final evaluation.

6. Main Goal

The goal of BADIRA is to build a **self-adaptive prediction framework** that can:

- Understand the quality of the available data.
- Select a suitable prediction model.
- Predict pregnancy complication risk.
- Provide a confidence score.
- Explain why the prediction is trustworthy.
- Work across different pregnancy complications.

The framework will be tested first on preeclampsia, followed by preterm birth and GDM, to evaluate whether BADIRA can adapt to different clinical prediction tasks.

7. References

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